INTRODUCTION / POSITIONING

Engineering data. Finding meaning.

Data Engineering / Applied Artificial Intelligence / Business Intelligence

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I build reliable data pipelines, explore machine-learning methods and translate complex datasets into clear, decision-ready analysis.

PROFILE

Master 2 Big Data, Analyse et Business Intelligence (BIDABI), Université Sorbonne Paris Nord. Previous Master in Information Systems and Software Engineering. Data Analyst internship at AGRODIV SPA.

FEATURED WORK

01 Customer Retention Analytics

Transaction processing, RFM segmentation and cohorts

02 DVF Île-de-France

Reliable real-estate ETL and analytical modeling

03 Intrusion Detection

CICIDS2017 machine-learning experimentation

Seeking a six-month internship in Île-de-France from January 2027.

CUSTOMER ANALYTICS

Customer Retention Analytics

From transaction records to a clearer picture of customer value.

1,067,371	779,425	5,878
Source rows	Clean sale rows	Customers

Problem

Customer purchases were distributed across transaction-level records with duplicates, cancellations and missing customer identifiers. Business analysis required a reliable analytical scope.

Solution

Designed a Python/pandas cleaning pipeline, created orders, RFM segments, monthly KPIs and cohort-retention outputs, and loaded analytical tables into PostgreSQL for SQL and BI use.

Impact & interpretation

The recorded dataset contains 36,969 orders. Champions: 1,473 customers contributing approximately 69% of analyzed positive sales (~£17.37M in total). These are analysis results on a public dataset, not employer revenue.

Data quality

Clean customer analysis excludes cancellations and anonymous transactions. The dashboard and screenshots should be checked in the project itself before linking to a public demo.

ARCHITECTURE

01	**Extract** Online Retail II source data
02	**Clean** Python validation and deduplication
03	**Model** Orders, RFM, cohorts and KPIs
04	**Report** PostgreSQL SQL and Power BI

TOOLS: Python · pandas · SQL · PostgreSQL · Docker · pytest · Power BI

github.com/samirmihoubi2003/customer-retention-analytics

DATA ENGINEERING

DVF Île-de-France Analytics

Transforming raw public real-estate files into analyzable transactions.

164,173	8	6
Transactions	Departments	pytest checks

Problem

DVF property files contain multiple rows per sale and data requiring rigorous validation. Naive aggregation can exaggerate transaction counts and distort price-per-square-metre analysis.

Solution

Built a Python ETL with PostgreSQL storage and a transaction-grain star-schema mart. Used Docker for reproducible local infrastructure and pytest, Ruff and GitHub Actions for quality checks.

Impact & interpretation

A recorded local run processed 164,173 transactions across eight Île-de-France departments. Counts are development-run results; they should be backed by reproducible run logs in the repository.

Analytical caution

Price comparisons depend on eligibility filters, property type and definition of transaction. No claim is made that the Power BI dashboard is published or production-ready.

ARCHITECTURE

01	**Source**	DVF open real-estate files
02	**Transform**	Checks and deduplication in Python
03	**Model**	PostgreSQL fact and dimensions
04	**Analyze**	SQL price/m² and geography

TOOLS: Python · SQL · PostgreSQL · Docker · pytest · Ruff · GitHub Actions

github.com/samirmihoubi2003/dvf-idf-analytics

APPLIED MACHINE LEARNING

Network Intrusion Detection

A methodical experimental comparison for network security analytics.

0.92	98.5%	CICIDS2017
Attack-class F1	Reported accuracy	Dataset

Problem

Intrusion detection requires identifying malicious patterns in heterogeneous, imbalanced network traffic. Train/test split methodology matters for valid evaluation.

Solution

Explored CICIDS2017 preprocessing and compared Isolation Forest, autoencoder and CNN-MLP approaches using confusion matrices, precision, recall, F1 and accuracy.

Impact & interpretation

Prior experimental records include attack-class F1 around 0.92 and accuracy around 98.5%. These originate from different reported conditions and must not be treated as a single unified benchmark.

Evaluation boundaries

Academic results, not deployed threat detection. Publish exact split, feature preprocessing and class definitions before treating the metrics as independently reproducible.

ARCHITECTURE

01	**Input**	CICIDS2017 network flows
02	**Process**	Clean, split and transform
03	**Model**	Isolation Forest · AE · CNN-MLP
04	**Evaluate**	Precision · Recall · F1

TOOLS: Python · scikit-learn · Isolation Forest · Autoencoder · CNN-MLP

github.com/samirmihoubi2003/memoire-ids-anomaly-detection

RESEARCH / EXPERIENCE / CONTACT

Beyond the featured work.

Supporting academic work, practical experience and ways to connect.

Word2Vec / NLP

Film descriptions and smartphone reviews; domain-specific embeddings and similarity analysis.

Poisson vs Negative Binomial

Overdispersed count data, 900 observations and information-criterion comparison.

Clustering & PCA

K-means, hierarchical clustering, silhouette evaluation and image segmentation.

PageRank & crawling

Directed graph ranking and university-site crawling experiments.

Classification & Apriori

Classifier comparison with cross-validation; association-rule mining using R.

PROFESSIONAL EXPERIENCE

Data Analyst Intern — AGRODIV SPA · Nov–Dec 2025

Three Power BI dashboards, eight operational KPIs, more than 15,000 cleaned records using SQL and Python, with approximately 25% less reporting preparation time.

CONTACT

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